

Can Market Signals Improve Delta Hedging?

A leakage-controlled replication experiment on historical SPY paths

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Quantitative Finance Project

2026-07-10

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Hi everyone. Today we will share our quantitative finance project on whether public market signals can improve delta hedging. In our leakage-controlled experiment on historical SPY paths, VIX reduces the largest hedging losses, while our more complex composite signal does not improve average accuracy.

We test hedge inputs – not historical option execution

Can point-in-time signals reduce terminal replication error versus fixed-volatility Black–Scholes?

Controlled comparison

Same payoff, initial premium, hedge calendar, and transaction-cost rule.

OBSERVED MARKET DATA

- ▶ SPY adjusted prices and volume
- ▶ VIX, XLK returns, and short rates

MODEL-DEFINED CLAIMS

- ▶ European calls with controlled maturity and moneyness
- ▶ No historical option bids, asks, or contract IV

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└ We test hedge inputs – not historical option execution

First, the scope: prices, volume, VIX, sector returns, and rates are real point-in-time observations. The options are standardized European calls placed on those paths; we do not use historical option quotes. Holding payoff, premium, hedge dates, and costs fixed lets us isolate the volatility input.

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Same payoff, initial premium, hedge calendar, and transaction-cost rule.

Four strategies; one common delta engine

FV-BS

Fixed 21-day realized volatility

Benchmark fixed at contract initiation.

ROLLING-BS

Updated 21-day realized volatility

Uses the latest price information.

VIX-BS

Daily VIX as hedge volatility

Adds forward-looking option-market information.

MSA-DELTA

Rolling vol $\times$ risk multiplier

Combines seven market signals.

Common premium · daily rebalancing · identical payoff · identical costs

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Four strategies; one common delta engine



The four strategies share one Black–Scholes delta engine. FV-BS freezes 21-day realized volatility; Rolling-BS updates it daily; VIX-BS uses daily VIX; and MSA-Delta adjusts rolling volatility with seven signals. Everything else is identical, so differences come from the volatility input.

Definition: volatility changes delta away from the money

COMMON HEDGE FORMULA

$$\Delta_t = N(d_{1,t}), \quad d_{1,t} = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma_t^2)\tau}{\sigma_t\sqrt{\tau}}.$$

Example contract

$S = 100$, $K = 105$, $\tau = 21/252$, and $r = 3\%$.

Why this matters: volatility adjustments materially change OTM and ITM deltas, but have little effect on an ATM delta near 0.5.

NORMAL STATE

$$\sigma = 20\% \Rightarrow \Delta = 0.220$$

+0.077 shares
↓
per option

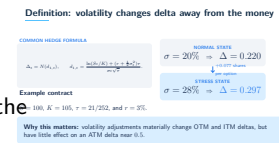
STRESS STATE

$$\sigma = 28\% \Rightarrow \Delta = 0.297$$

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Definition: volatility changes delta away from the money



Why does the input matter? In this out-of-the-money example, raising volatility from 20 to 28 percent raises delta from 0.220 to 0.297—about eight additional shares per hundred options. The effect is strongest away from the money; at-the-money delta remains near one-half.

Definition: seven trailing signals create three risk states

STANDARDIZED RISK SCORE

$$R_t = 0.30 \text{IVRank}_t + 0.20 \text{VIX}_t + 0.20 \frac{\text{RV}_{21,t}}{\text{RV}_{63,t}} + 0.10 \text{VolumeShock}_t + 0.10 \text{Drawdown}_{63,t} + 0.05 \text{SectorDiv}_t + 0.05 |\Delta \text{Rate}_{21,t}|.$$

Features are standardized using development-period statistics only.

$$\text{NORMAL} \quad R_t < q_{70} \quad m = 1.00$$

$$\text{ELEVATED} \quad q_{70} \leq R_t < q_{90} \\ m = 1.15$$

$$\text{STRESS} \quad R_t \geq q_{90} \quad m = 1.40$$

$$\sigma_t^{\text{MSA}} = m(Z_t) \hat{\sigma}_{21,t}$$

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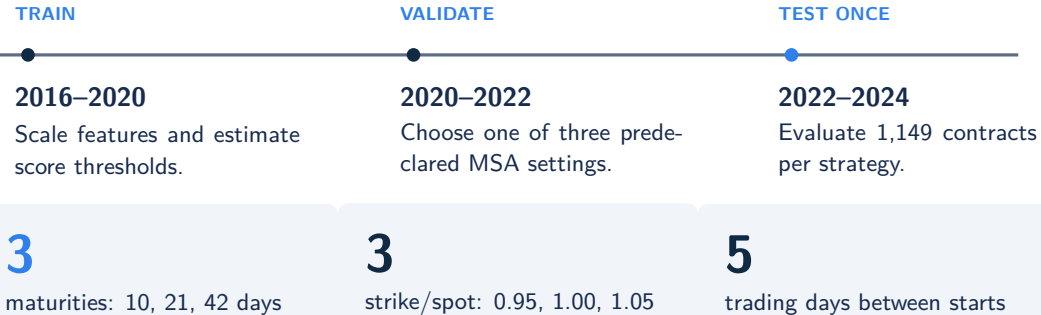
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Definition: seven trailing signals create three risk states



Our composite score combines seven trailing indicators, led by VIX rank, VIX level, and the realized-volatility ratio. Development-period statistics standardize the features. The score maps to normal, elevated, or stress multipliers of 1, 1.15, and 1.4. Every day- t input was observable on day t .

The test period stays untouched until final evaluation



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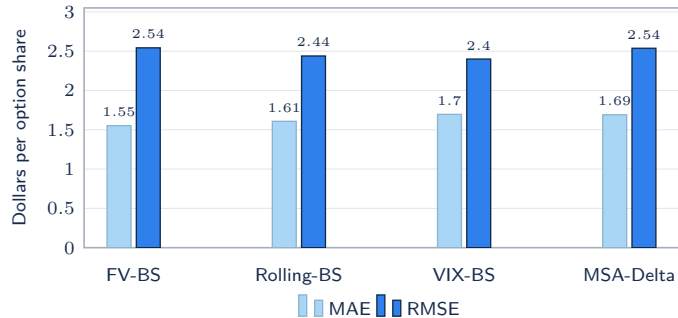
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The test period stays untouched until final evaluation



This split protects against leakage. We fit scaling and thresholds in 2016–2020, select one predeclared setting in 2020–2022, and evaluate once on untouched 2022–2024 data. Three maturities and three moneyness levels, starting every five days, produce 1,149 contracts per strategy.

No single hedge wins every error metric



1.552

Lowest MAE: FV-BS

2.399

Lowest RMSE: VIX-BS

Interpretation

Average accuracy favors simplicity; large-error control favors forward-looking volatility.

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No single hedge wins every error metric



Here are the headline test results; lower is better. FV-BS has the lowest MAE at 1.552, so it wins on average absolute accuracy. VIX-BS has the lowest RMSE at 2.399, meaning it better controls large errors. MSA-Delta wins neither metric.

VIX helps most where losses are largest

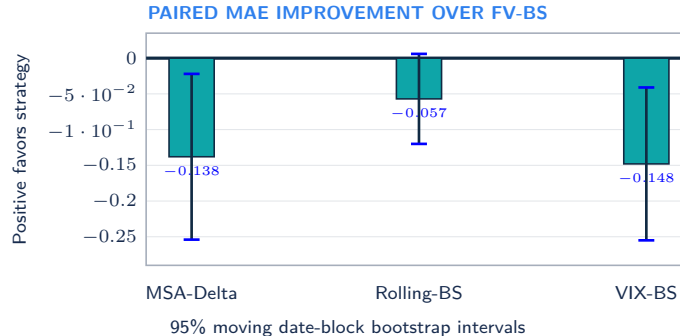
5.52

VIX-BS 95% loss Expected Shortfall

6.75

FV-BS 95% loss Expected Shortfall

Lower is better. VIX reduces expected loss in the worst 5% of outcomes by about **18%**.



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VIX helps most where losses are largest



The benefit appears in the loss tail. In the worst five percent of outcomes, VIX-BS reduces expected loss from 6.75 to 5.52 dollars—about 18 percent. But every MAE-improvement estimate is below zero; the bootstrap intervals show no average-error advantage for VIX or MSA. VIX helps severe losses, not typical absolute error.

The ranking survives transaction-cost changes

Transaction cost	Lowest MAE		Lowest RMSE	
0 bps	FV-BS	1.456	VIX-BS	2.394
5 bps	FV-BS	1.552	VIX-BS	2.399
10 bps	FV-BS	1.745	VIX-BS	2.510

STABLE FINDING 1

Fixed volatility remains best for average absolute error.

STABLE FINDING 2

VIX remains best for squared error at every cost level.

Results are also reported for all nine maturity–moneyness cells.

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The ranking is stable at zero, five, and ten basis points: FV-BS retains the lowest MAE, while VIX-BS retains the lowest RMSE. Winners vary across individual maturity and moneyness cells, so no method dominates every contract and objective.

The useful signal is simpler than the model we designed

Forward-looking volatility improves tail-risk control – but more signals do not guarantee a better hedge.

KEEP

VIX-BS as the strongest tail-risk benchmark.

REJECT

A blanket claim that additional complexity improves hedging.

NEXT

Repeat the design using contract-level option quotes when available.

IMPORTANT LIMITATIONS

- ▶ VIX is an index-level 30-day proxy, not contract-level SPY IV.
- ▶ Controlled European claims omit historical spreads and early exercise.
- ▶ Block bootstrap mitigates, but does not eliminate, overlap dependence.

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Our conclusion is simple: keep VIX-BS as a strong tail-risk benchmark, but reject the claim that more signals automatically improve hedging. The main limits are VIX as a proxy, standardized European claims, and overlapping contracts. Next, we would repeat the design with contract-level option quotes. Thank you—we welcome your questions.